

PMM Messaging Engine

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PMM Messaging Engine — Comprehensive Documentation

1. Executive Summary

The PMM Messaging Engine is a full-stack application that generates, scores, and refines marketing messaging assets from product documentation and community evidence. Users create workspace sessions, select asset types and voice profiles, choose a generation pipeline, and iteratively refine the output through actions and chat.

Core value proposition: Every messaging asset is grounded in real community evidence, enriched with competitive intelligence, generated in a controlled voice, stress-tested for quality, and fully traceable back to its source.

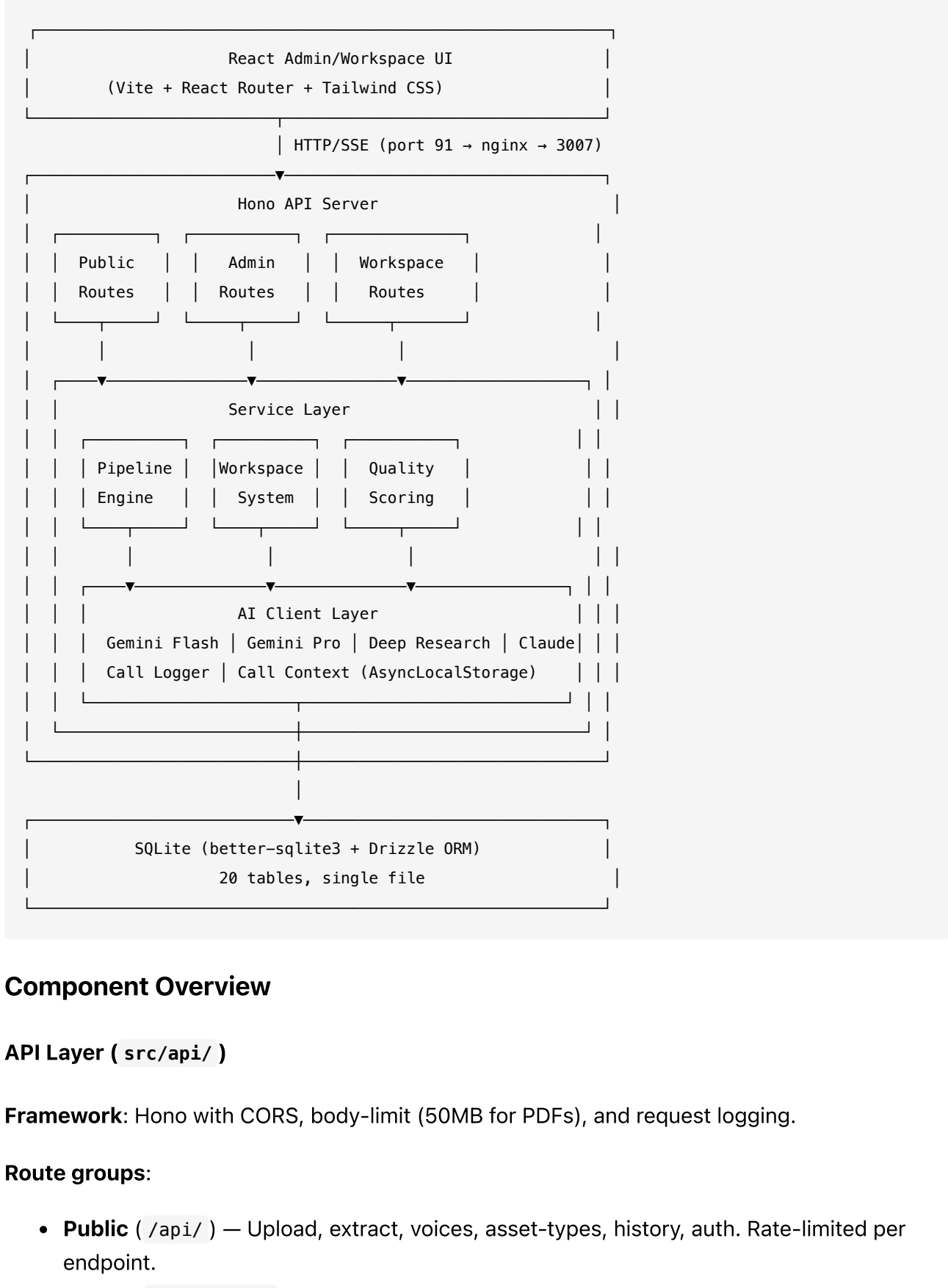
Tech Stack

Component	Technology
Runtime	Node.js + TypeScript (ESM)
Backend	Hono (lightweight web framework)
Database	SQLite via better-sqlite3 + Drizzle ORM
Frontend	Vite + React + React Router + Tailwind CSS
AI Models	Gemini Flash, Gemini Pro, Deep Research, Claude (optional)
Auth	JWT via jose, bcryptjs for passwords
IDs	nanoid (21-character strings)
Process Manager	PM2
Testing	Vitest (5min timeout per test)

2. System Architecture



System Architecture



Component Overview

API Layer (`src/api/`)

Framework: Hono with CORS, body-limit (50MB for PDFs), and request logging.

Route groups:

- **Public** (`/api/`) — Upload, extract, voices, asset-types, history, auth. Rate-limited per endpoint.
- **Admin** (`/api/admin/`) — Documents, voices, settings, stats. JWT auth required.
- **Workspace** (`/api/workspace/`) — Sessions, versions, actions, chat. JWT auth required (user-scoped).

Auth: JWT via jose library. Two auth middlewares:

- `adminAuth` — for admin routes (accepts env var credentials or DB users with admin role)
- `workspaceAuth` — for workspace routes (accepts any authenticated user)

Service Layer

The service layer contains the core business logic:

- **Pipeline Engine** — 5 generation pipelines sharing common orchestrator primitives
- **Workspace System** — Session management, versioning, chat, background actions
- **Quality Scoring** — 5-dimension parallel scoring with quality gates
- **Product Insights** — Extraction and tiered formatting of product documentation
- **Evidence & Research** — Community and competitive deep research

AI Client Layer (`src/services/ai/`)

Unified client layer with:

- `generateWithClaude()` — Claude API with rate limiting + retry
- `generateWithGemini()` — Gemini with Flash/Pro routing + rate limiting + retry
- `generateWithGeminiGroundedSearch()` — Gemini with Google Search tool (5x empty retry)
- `generateJSON<T>()` — Gemini Pro JSON generation with parse retry
- `createDeepResearchInteraction()` — Async deep research submission
- `pollInteractionUntilComplete()` — Poll with configurable interval/timeout

Rate limiters: Claude (10/min), Gemini Flash (60/min), Gemini Pro (15/min)

Model Profile System

Controlled by `MODEL_PROFILE` env var (`'economy'` | `'premium'`).

Task	Premium Model	Economy Model
flash	gemini-flash-latest	gemini-flash-lite-latest
pro	gemini-pro-latest	gemini-flash-lite-latest
deepResearch	deep-research-max-preview-04-2026	gemini-flash-lite-latest
generation	gemini-pro-latest	gemini-flash-lite-latest
scoring	gemini-flash-latest	gemini-flash-lite-latest
deslop	gemini-pro-latest	gemini-flash-lite-latest

Use `getModelForTask(task)` from `config.ts` — never hardcode model names.

Deployment

- **Server:** Hetzner VPS at 5.161.203.108
- **Process:** PM2 via `ecosystem.config.cjs`
- **Port:** 3007 (app) → 91 (nginx proxy)
- **Deploy:** `./deploy.sh` (npm install → build → admin build → git commit → PM2 restart)
- **Database:** `data/messaging-engine.db` (SQLite, single file)

3. Database Schema

SQLite database via better-sqlite3 + Drizzle ORM. Schema defined in `src/db/schema.ts`. **20 tables total.**

Table Overview

#	Table	Purpose
1	<code>messaging_priorities</code>	Strategic messaging themes/niches
2	<code>discovery_schedules</code>	Source polling configuration
3	<code>discovered_pain_points</code>	Pain points from community sources
4	<code>generation_jobs</code>	Generation pipeline job tracking
5	<code>settings</code>	Key-value configuration
6	<code>product_documents</code>	Uploaded product context docs
7	<code>messaging_assets</code>	Generated messaging assets (primary output)
8	<code>persona_critics</code>	AI critic personas for scoring
9	<code>persona_scores</code>	Per-asset, per-critic scores
10	<code>competitive_research</code>	Deep Research results
11	<code>asset_traceability</code>	Evidence chain per asset
12	<code>messaging_gaps</code>	Identified coverage gaps
13	<code>voice_profiles</code>	Voice/tone profiles
14	<code>asset_variants</code>	Per-voice asset variants with scores
15	<code>users</code>	Workspace user accounts
16	<code>sessions</code>	Workspace sessions
17	<code>session_versions</code>	Versioned asset content per session
18	<code>session_messages</code>	Chat refinement messages
19	<code>action_jobs</code>	Async workspace action tracking
20	<code>llm_calls</code>	LLM call audit log

Table 1: `messaging_priorities`

Strategic messaging themes that group pain points and assets.

Column	Type	Constraints	Description
<code>id</code>	TEXT	PK	nanoid
<code>name</code>	TEXT	NOT NULL	Display name
<code>slug</code>	TEXT	NOT NULL, UNIQUE	URL-safe identifier
<code>description</code>	TEXT	NOT NULL	Theme description
<code>keywords</code>	TEXT	NOT NULL	JSON array of keywords
<code>product_context</code>	TEXT	NOT NULL	Product context for this priority
<code>is_active</code>	INTEGER	NOT NULL, default true	Active flag
<code>created_at</code>	TEXT	NOT NULL	ISO 8601
<code>updated_at</code>	TEXT	NOT NULL	ISO 8601

Table 2: `discovery_schedules`

Configuration for automated source polling.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
priority_id	TEXT	NOT NULL, FK → messaging_priorities	Parent priority
source_type	TEXT	NOT NULL	Source type identifier
config	TEXT	NOT NULL	JSON configuration
is_active	INTEGER	NOT NULL, default true	Active flag
last_run_at	TEXT	nullable	Last execution time
next_run_at	TEXT	nullable	Scheduled next run
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 3: discovered_pain_points

Pain points extracted from community sources or manually entered.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
priority_id	TEXT	NOT NULL, FK → messaging_priorities	Parent priority
schedule_id	TEXT	FK → discovery_schedules	Source schedule
source_type	TEXT	NOT NULL	'reddit', 'hn', 'manual', etc.
source_url	TEXT	NOT NULL	Original URL
source_id	TEXT	NOT NULL	External identifier
title	TEXT	NOT NULL	Pain point title
content	TEXT	NOT NULL	Full content
author	TEXT	NOT NULL	Author name
author_level	TEXT	NOT NULL	Author expertise level
metadata	TEXT	NOT NULL	JSON metadata
pain_score	REAL	NOT NULL	Pain severity (0–1)
pain_analysis	TEXT	NOT NULL	JSON analysis
practitioner_quotes	TEXT	NOT NULL	JSON array of raw quotes
status	TEXT	NOT NULL, default 'pending'	'pending', 'approved', 'rejected'
rejection_reason	TEXT	nullable	Why rejected
content_hash	TEXT	NOT NULL	Dedup hash
discovered_at	TEXT	NOT NULL	Discovery timestamp
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 4: generation_jobs

Tracks generation pipeline execution.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
pain_point_id	TEXT	FK → discovered_pain_points	Source pain point
priority_id	TEXT	FK → messaging_priorities	Priority context
status	TEXT	NOT NULL, default 'pending'	'pending', 'running', 'completed', 'failed'
current_step	TEXT	nullable	Human-readable progress step
progress	INTEGER	NOT NULL, default 0	Progress percentage (0–100)
competitive_research	TEXT	nullable	JSON research results
product_context	TEXT	nullable	JSON: productDocs, voiceProfile, fields, assetTypes, model, pipeline
error_message	TEXT	nullable	Error description
error_stack	TEXT	nullable	Error stack trace
retry_count	INTEGER	NOT NULL, default 0	Retry attempts
gemini_interaction_id	TEXT	nullable	Deep Research interaction ID
gemini_status	TEXT	nullable	Deep Research status
started_at	TEXT	nullable	Start timestamp
pipeline_steps	TEXT	nullable	JSON array of pipeline step events
completed_at	TEXT	nullable	Completion timestamp
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 5: settings

Key-value configuration store.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
priority_id	TEXT	FK → messaging_priorities	Scoped to priority (optional)
key	TEXT	NOT NULL	Setting key
value	TEXT	NOT NULL	Setting value
description	TEXT	NOT NULL	Human description
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 6: product_documents

Uploaded product documentation used as generation context.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
name	TEXT	NOT NULL	Document name
description	TEXT	NOT NULL	Description
content	TEXT	NOT NULL	Full text content
document_type	TEXT	NOT NULL	Type classification
tags	TEXT	NOT NULL	JSON array of tags
is_active	INTEGER	NOT NULL, default true	Active flag
uploaded_at	TEXT	NOT NULL	Upload timestamp
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 7: messaging_assets

Generated messaging assets — the primary output of the system.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
priority_id	TEXT	NOT NULL, FK → messaging_priorities	Priority context
job_id	TEXT	FK → generation_jobs	Source generation job
pain_point_id	TEXT	FK → discovered_pain_points	Source pain point
asset_type	TEXT	NOT NULL	Asset type (8 types)
title	TEXT	NOT NULL	Asset title
content	TEXT	NOT NULL	Full generated content
metadata	TEXT	NOT NULL	JSON: generationId, voiceId, voiceName, etc.
slop_score	REAL	nullable	Slop score (0–10, lower is better)
vendor_speak_score	REAL	nullable	Vendor–speak score (0–10, lower is better)
specificity_score	REAL	nullable	Specificity score (0–10, higher is better)
persona_avg_score	REAL	nullable	Persona average (0–10, higher is better)
evidence_level	TEXT	nullable	'strong', 'partial', 'product-only'
status	TEXT	NOT NULL, default 'draft'	'draft', 'review', 'approved'
review_notes	TEXT	nullable	Reviewer notes
approved_at	TEXT	nullable	Approval timestamp
approved_by	TEXT	nullable	Approver
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 8: persona_critics

AI critic personas used for quality scoring.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
name	TEXT	NOT NULL	Critic name
description	TEXT	NOT NULL	Role/perspective description
prompt_template	TEXT	NOT NULL	Scoring prompt template
is_active	INTEGER	NOT NULL, default true	Active flag
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 9: persona_scores

Individual scores from each critic for each asset.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
asset_id	TEXT	NOT NULL, FK → messaging_assets	Scored asset
persona_id	TEXT	NOT NULL, FK → persona_critics	Scoring critic
score	REAL	NOT NULL	Score value
feedback	TEXT	NOT NULL	Textual feedback
strengths	TEXT	NOT NULL	JSON array
weaknesses	TEXT	NOT NULL	JSON array
created_at	TEXT	NOT NULL	ISO 8601

Table 10: competitive_research

Stored results from Gemini Deep Research competitive analysis.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
job_id	TEXT	NOT NULL, FK → generation_jobs	Parent job
pain_point_id	TEXT	NOT NULL, FK → discovered_pain_points	Related pain point
raw_report	TEXT	NOT NULL	Full research report
structured_analysis	TEXT	NOT NULL	JSON structured analysis
grounding_sources	TEXT	NOT NULL	JSON source URLs
gemini_interaction_id	TEXT	NOT NULL	Deep Research ID
status	TEXT	NOT NULL, default 'pending'	Research status
created_at	TEXT	NOT NULL	ISO 8601

Table 11: asset_traceability

Complete evidence chain for each generated asset.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
asset_id	TEXT	NOT NULL, FK → messaging_assets	Traced asset
pain_point_id	TEXT	FK → discovered_pain_points	Source pain point
research_id	TEXT	FK → competitive_research	Source research
product_doc_id	TEXT	FK → product_documents	Source document
practitioner_quotes	TEXT	NOT NULL	JSON array of quotes
generation_prompt	TEXT	nullable	JSON: {system, user, timestamp}
created_at	TEXT	NOT NULL	ISO 8601

Table 12: messaging_gaps

Identified gaps in messaging coverage.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
pain_point_id	TEXT	FK → discovered_pain_points	Related pain point
description	TEXT	NOT NULL	Gap description
suggested_capability	TEXT	NOT NULL	Suggested capability
frequency	INTEGER	NOT NULL, default 1	Occurrence count
status	TEXT	NOT NULL, default 'open'	'open', 'addressed'
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 13: voice_profiles

Voice and tone profiles for generation and quality scoring.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
name	TEXT	NOT NULL	Profile name
slug	TEXT	NOT NULL, UNIQUE	URL-safe slug
description	TEXT	NOT NULL	Profile description
voice_guide	TEXT	NOT NULL	Full voice guide text
scoring_thresholds	TEXT	NOT NULL	JSON: {slopMax, vendorSpeakMax, authenticityMin, specificityMin, personaMin}
example_phrases	TEXT	NOT NULL	JSON array of example phrases
is_default	INTEGER	NOT NULL, default false	Default profile flag
is_active	INTEGER	NOT NULL, default true	Active flag
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 14: asset_variants

Per-voice variants of messaging assets with full quality scores.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
asset_id	TEXT	NOT NULL, FK → messaging_assets	Parent asset
voice_profile_id	TEXT	NOT NULL, FK → voice_profiles	Voice used
variant_number	INTEGER	NOT NULL	Variant sequence number
content	TEXT	NOT NULL	Variant content
slop_score	REAL	nullable	Slop score
vendor_speak_score	REAL	nullable	Vendor–speak score
authenticity_score	REAL	nullable	Authenticity score
specificity_score	REAL	nullable	Specificity score
persona_avg_score	REAL	nullable	Persona average score
passes_gates	INTEGER	NOT NULL, default false	Quality gate result
is_selected	INTEGER	NOT NULL, default false	Selected variant flag
created_at	TEXT	NOT NULL	ISO 8601

Table 15: users

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
name	TEXT	NOT NULL	Profile name
slug	TEXT	NOT NULL, UNIQUE	URL-safe slug
description	TEXT	NOT NULL	Profile description
voice_guide	TEXT	NOT NULL	Full voice guide text
scoring_thresholds	TEXT	NOT NULL	JSON: {slopMax, vendorSpeakMax, authenticityMin, specificityMin, personaMin}
example_phrases	TEXT	NOT NULL	JSON array of example phrases
is_default	INTEGER	NOT NULL, default false	Default profile flag
is_active	INTEGER	NOT NULL, default true	Active flag
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 16: user_preferences

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
user_id	TEXT	NOT NULL, FK → users	User
voice_profile_id	TEXT	NOT NULL, FK → voice_profiles	Voice profile
variant_number	INTEGER	NOT NULL	Variant sequence number
content	TEXT	NOT NULL	Variant content
slop_score	REAL	nullable	Slop score
vendor_speak_score	REAL	nullable	Vendor–speak score
authenticity_score	REAL	nullable	Authenticity score
specificity_score	REAL	nullable	Specificity score
persona_avg_score	REAL	nullable	Persona average score
passes_gates	INTEGER	NOT NULL, default false	Quality gate result
is_selected	INTEGER	NOT NULL, default false	Selected variant flag
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 17: system_logs

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
component	TEXT	NOT NULL	System component
message	TEXT	NOT NULL	Log message
severity	TEXT	NOT NULL	Log severity level
context	TEXT	NOT NULL	JSON context data
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 18: audit_logs

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
actor_id	TEXT	NOT NULL, FK → users	User performing action
target_id	TEXT	NOT NULL, FK → assets	Targeted asset
action	TEXT	NOT NULL	Action performed
details	TEXT	NOT NULL	JSON details of action
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 19: deployment_history

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
version	TEXT	NOT NULL	Deployment version
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
environment	TEXT	NOT NULL	Deployment environment
status	TEXT	NOT NULL	Deployment status
message	TEXT	NOT NULL	Deployment message
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 20: feature_flags

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
name	TEXT	NOT NULL	Feature flag name
description	TEXT	NOT NULL	Feature flag description
status	TEXT	NOT NULL	Feature flag status
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 21: analytics_events

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
event_type	TEXT	NOT NULL	Analytics event type
user_id	TEXT	NOT NULL, FK → users	User ID
session_id	TEXT	NOT NULL, FK → sessions	Session ID
properties	TEXT	NOT NULL	JSON event properties
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 22: session_data

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
user_id	TEXT	NOT NULL, FK → users	User ID
start_time	TEXT	NOT NULL	Session start time (ISO 8601)
end_time	TEXT	NOT NULL	Session end time (ISO 8601)
duration	TEXT	NOT NULL	Session duration
events	TEXT	NOT NULL	JSON array of events
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 23: user_sessions

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
user_id	TEXT	NOT NULL, FK → users	User ID
session_id	TEXT	NOT NULL, FK → session_data	Session ID
start_time	TEXT	NOT NULL	Session start time (ISO 8601)
end_time	TEXT	NOT NULL	Session end time (ISO 8601)
duration	TEXT	NOT NULL	Session duration
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 24: user_preferences

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
user_id	TEXT	NOT NULL, FK → users	User ID
voice_profile_id	TEXT	NOT NULL, FK → voice_profiles	Voice profile
variant_number	INTEGER	NOT NULL	Variant sequence number
content	TEXT	NOT NULL	Variant content
slop_score	REAL	nullable	Slop score
vendor_speak_score	REAL	nullable	Vendor–speak score
authenticity_score	REAL	nullable	Authenticity score
specificity_score	REAL	nullable	Specificity score
persona_avg_score	REAL	nullable	Persona average score
passes_gates	INTEGER	NOT NULL, default false	Quality gate result
is_selected	INTEGER	NOT NULL, default false	Selected variant flag
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 25: system_logs

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
component	TEXT	NOT NULL	System component
message	TEXT	NOT NULL	Log message
severity	TEXT	NOT NULL	Log severity level
context	TEXT	NOT NULL	JSON context data
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 26: audit_logs

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
actor_id	TEXT	NOT NULL, FK → users	User performing action
target_id	TEXT	NOT NULL, FK → assets	Targeted asset
action	TEXT	NOT NULL	Action performed
details	TEXT	NOT NULL	JSON details of action
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 27: deployment_history

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
version	TEXT	NOT NULL	Deployment version
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
environment	TEXT	NOT NULL	Deployment environment
status	TEXT	NOT NULL	Deployment status
message	TEXT	NOT NULL	Deployment message
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 28: feature_flags

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
name	TEXT	NOT NULL	Feature flag name
description	TEXT	NOT NULL	Feature flag description
status	TEXT	NOT NULL	Feature flag status
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 29: analytics_events

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
timestamp	TEXT	NOT NULL	ISO 8601 timestamp
event_type	TEXT	NOT NULL	Analytics event type
user_id	TEXT	NOT NULL, FK → users	User ID
session_id	TEXT	NOT NULL, FK → sessions	Session ID

Workspace user accounts.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
username	TEXT	NOT NULL, UNIQUE	Login username
email	TEXT	NOT NULL, UNIQUE	Email address
password_hash	TEXT	NOT NULL	bcrypt hash (12 rounds)
display_name	TEXT	NOT NULL	Display name
role	TEXT	NOT NULL, default 'user'	'user' or 'admin'
is_active	INTEGER	NOT NULL, default true	Active flag
last_login_at	TEXT	nullable	Last login timestamp
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

First registered user automatically gets `admin` role.

Table 16: sessions

Workspace sessions — the primary workspace unit.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
user_id	TEXT	NOT NULL, FK → users	Owner
name	TEXT	NOT NULL	Session name (auto-generated from insights)
pain_point_id	TEXT	FK → discovered_pain_points	Linked pain point
job_id	TEXT	FK → generation_jobs	Generation job
voice_profile_id	TEXT	FK → voice_profiles	Primary voice profile
asset_types	TEXT	NOT NULL	JSON array of AssetType strings
status	TEXT	NOT NULL, default 'pending'	'pending', 'generating', 'completed', 'failed'
manual_pain_point	TEXT	nullable	JSON: {title, description, quotes?}
product_doc_ids	TEXT	nullable	JSON array of product_documents IDs
product_context	TEXT	nullable	Pasted/uploaded text context
focus_instructions	TEXT	nullable	User focus/instructions
pipeline	TEXT	default 'outside-in'	Pipeline selection
metadata	TEXT	default '{}'	JSON: voiceProfileIds, existingMessaging, modelProfile
is_archived	INTEGER	NOT NULL, default false	Archive flag
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Multi-voice support via `metadata.voiceProfileIds` array. Single voice via `voice_profile_id` column.

Table 17: session_versions

Versioned asset content per session. Every change creates a new version.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
session_id	TEXT	NOT NULL, FK → sessions	Parent session
asset_type	TEXT	NOT NULL	Asset type
version_number	INTEGER	NOT NULL	Sequential version number
content	TEXT	NOT NULL	Version content
source	TEXT	NOT NULL	How created: 'generation', 'edit', 'deslop', 'regenerate', 'voice_change', 'adversarial', 'chat', 'competitive_dive', 'community_check', 'multi_perspective'
source_detail	TEXT	nullable	JSON context about trigger
slop_score	REAL	nullable	Slop score
vendor_speak_score	REAL	nullable	Vendor-speak score
authenticity_score	REAL	nullable	Authenticity score
specificity_score	REAL	nullable	Specificity score
persona_avg_score	REAL	nullable	Persona average score
passes_gates	INTEGER	NOT NULL, default false	Quality gate result
is_active	INTEGER	NOT NULL, default false	Currently active version
created_at	TEXT	NOT NULL	ISO 8601

Table 18: session_messages

Chat refinement message history per session.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
session_id	TEXT	NOT NULL, FK → sessions	Parent session
role	TEXT	NOT NULL	'user' or 'assistant'
content	TEXT	NOT NULL	Message text
asset_type	TEXT	nullable	Which asset tab was focused
version_created	TEXT	nullable	Version ID if content accepted
metadata	TEXT	default '{}'	JSON: token usage, model, latency
created_at	TEXT	NOT NULL	ISO 8601

Table 19: action_jobs

Async background workspace actions with progress tracking.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
session_id	TEXT	NOT NULL, FK → sessions	Parent session
asset_type	TEXT	NOT NULL	Target asset type
action_name	TEXT	NOT NULL	Action identifier
status	TEXT	NOT NULL, default 'pending'	'pending', 'running', 'completed', 'failed'
current_step	TEXT	nullable	Current progress step
progress	INTEGER	NOT NULL, default 0	Progress percentage
result	TEXT	nullable	JSON ActionResult on completion
error_message	TEXT	nullable	Error message on failure
created_at	TEXT	NOT NULL	ISO 8601
updated_at	TEXT	NOT NULL	ISO 8601

Table 20: llm_calls

Audit log of every LLM call made by the system.

Column	Type	Constraints	Description
id	TEXT	PK	nanoid
session_id	TEXT	FK → sessions	Linked workspace session
job_id	TEXT	FK → generation_jobs	Linked generation job
timestamp	TEXT	NOT NULL	Call timestamp
model	TEXT	NOT NULL	Model used
purpose	TEXT	NOT NULL	Call purpose
system_prompt	TEXT	nullable	System prompt sent
user_prompt	TEXT	NOT NULL	User prompt sent
response	TEXT	nullable	Model response
input_tokens	INTEGER	NOT NULL, default 0	Input token count
output_tokens	INTEGER	NOT NULL, default 0	Output token count
total_tokens	INTEGER	NOT NULL, default 0	Total tokens
cached_tokens	INTEGER	NOT NULL, default 0	Cached token count
latency_ms	INTEGER	NOT NULL, default 0	Response latency
success	INTEGER	NOT NULL, default true	Success flag
error_message	TEXT	nullable	Error message on failure
finish_reason	TEXT	nullable	Model finish reason
created_at	TEXT	NOT NULL	ISO 8601

Relationships

messaging_priorities	←<	discovery_schedules
messaging_priorities	←<	discovered_pain_points
messaging_priorities	←<	generation_jobs
messaging_priorities	←<	messaging_assets
messaging_priorities	←<	settings

discovered_pain_points	←<	generation_jobs
discovered_pain_points	←<	messaging_assets
discovered_pain_points	←<	competitive_research
discovered_pain_points	←<	messaging_gaps
discovered_pain_points	←<	asset_traceability

generation_jobs	←<	competitive_research
generation_jobs	←<	sessions (via job_id)
generation_jobs	←<	llm_calls (via job_id)

messaging_assets	←<	asset_variants
messaging_assets	←<	persona_scores
messaging_assets	←<	asset_traceability

voice_profiles	←<	asset_variants
voice_profiles	←<	sessions (via voice_profile_id)

users	←<	sessions
sessions	←<	session_versions
sessions	←<	session_messages
sessions	←<	action_jobs
sessions	←<	llm_calls (via session_id)

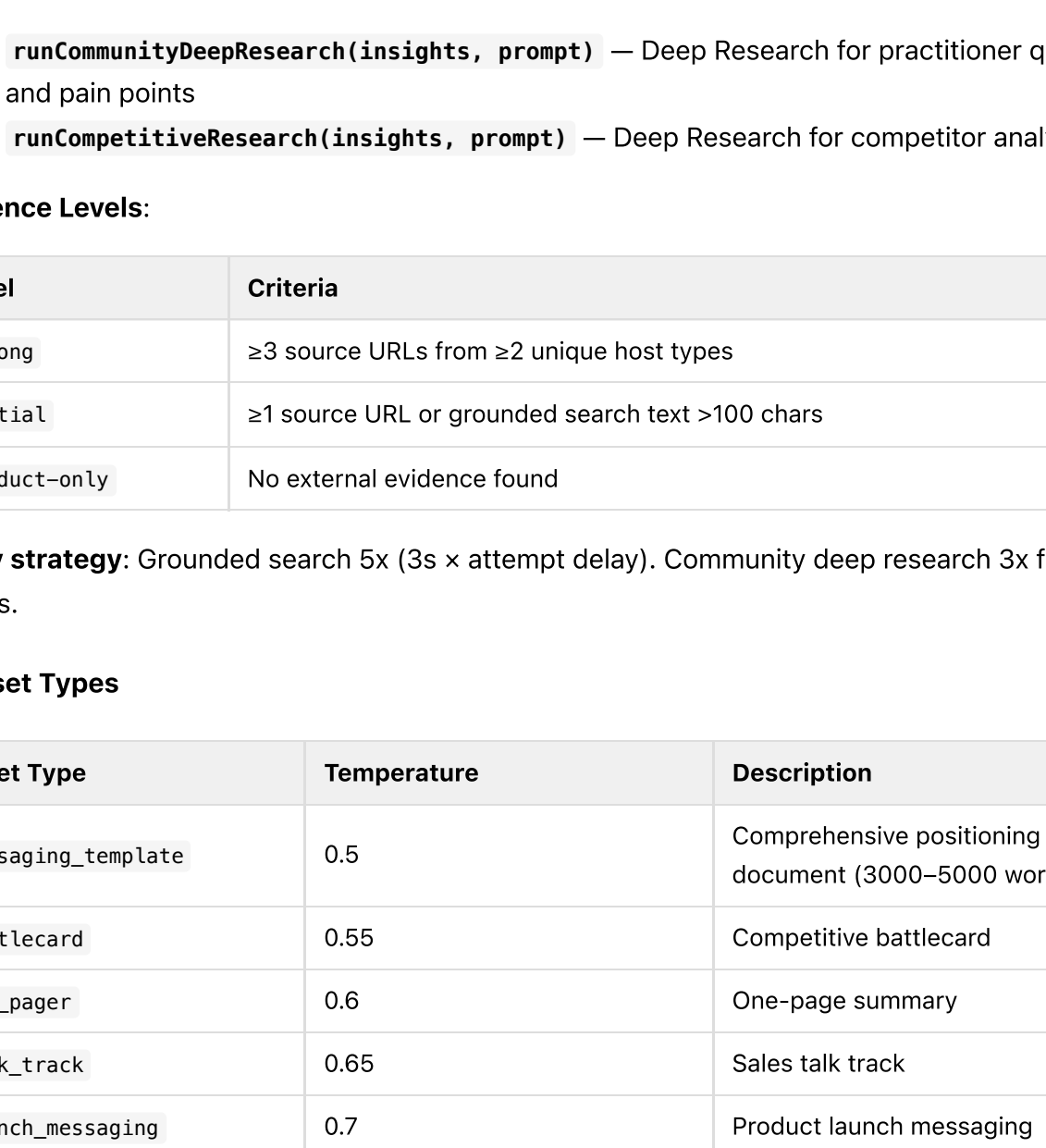
persona_critics	←<	persona_scores
product_documents	←<	asset_traceability

Database Conventions

- **IDs:** 21-character nanoid strings via `generateId()`
- **Timestamps:** ISO 8601 TEXT columns, never Unix timestamps
- **JSON columns:** Stored as TEXT, parsed with `JSON.parse()`, serialized with `JSON.stringify()`
- **Boolean columns:** SQLite INTEGER with `{ mode: 'boolean' }` — stores 0/1
- **Cascading deletes:** Most child tables CASCADE on parent delete
- **SET NULL:** Used for optional references that should survive parent deletion

4. Pipeline Architecture

The messaging engine has **5 generation pipelines**, each with different strategies for producing messaging assets. All pipelines share common primitives from `orchestrator.ts` and produce the same output format: scored messaging assets with full traceability.



All Pipelines Comparison

Pipeline Selection

Pipeline	Slug	Best For	Key Characteristic
Standard	standard	General messaging from product docs	Deep PoV extraction → research → generation
Outside-In	outside-in	Practitioner-authentic messaging	Community-first; fails if no evidence
Adversarial	adversarial	Battle-tested messaging	2 rounds of hostile critique + defense
Multi-Perspective	multi-perspective	Well-rounded messaging	3 angles synthesized into best version
Straight-Through	straight-through	Scoring existing content	No generation — score only

Shared Infrastructure

Orchestrator Primitives (`orchestrator.ts`)

Function	Purpose
loadJobInputs(jobId)	Load job configuration (product docs, voices, asset types, pipeline)
generateContent(prompt, options, model?)	AI dispatch — routes to Gemini or Claude
generateAndScore(prompt, system, model, context, thresholds, assetType)	Generate + score in one call
refinementLoop(content, context, thresholds, voice, assetType, system, model, maxIter, productName)	Iterative: deslop → refine → score until gates pass or plateau
storeVariant(jobId, assetType, voice, content, scores, passesGates, ...)	Store asset + variant + traceability records
emitPipelineStep(jobId, step, status, data?)	Record pipeline step events
updateJobProgress(jobId, fields)	Update job progress/status
finalizeJob(jobId, researchAvailable, researchLength)	Mark job complete

Evidence & Research (`evidence.ts`)

- `runCommunityDeepResearch(insights, prompt)` — Deep Research for practitioner quotes and pain points
- `runCompetitiveResearch(insights, prompt)` — Deep Research for competitor analysis

Evidence Levels:

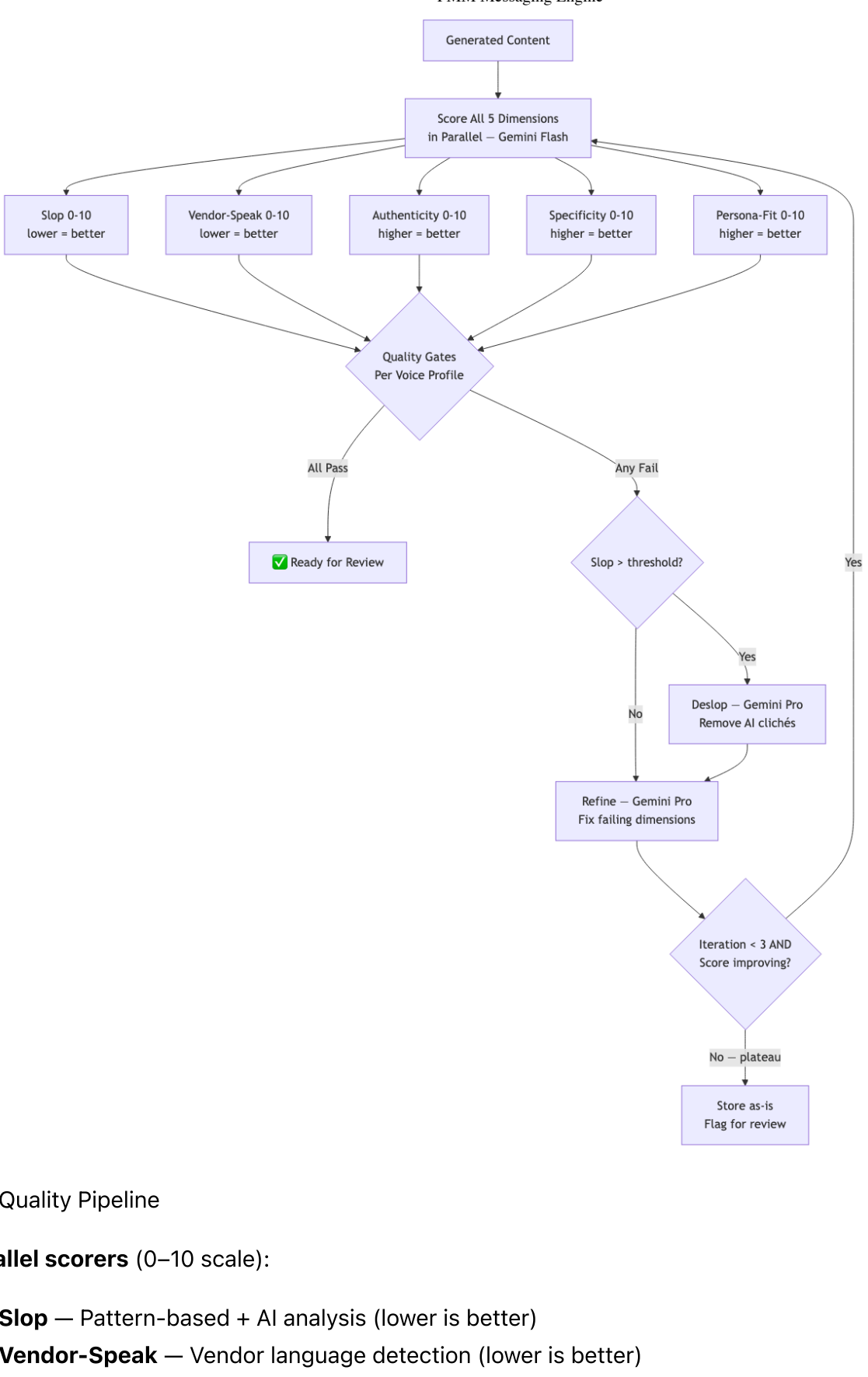
Level	Criteria
strong	≥3 source URLs from ≥2 unique host types
partial	≥1 source URL or grounded search text >100 chars
product-only	No external evidence found

Retry strategy: Grounded search 5x (3s × attempt delay). Community deep research 3x full retries.

8 Asset Types

Asset Type	Temperature	Description
messaging_template	0.5	Comprehensive positioning document (3000–5000 words)
battlecard	0.55	Competitive battlecard
one_pager	0.6	One-page summary
talk_track	0.65	Sales talk track
launch_messaging	0.7	Product launch messaging
email_copy	0.75	Email campaign copy
narrative	0.8	3-variant storytelling narrative
social_hook	0.85	Social media hooks

Quality Scoring



Quality Pipeline

5 parallel scorers (0–10 scale):

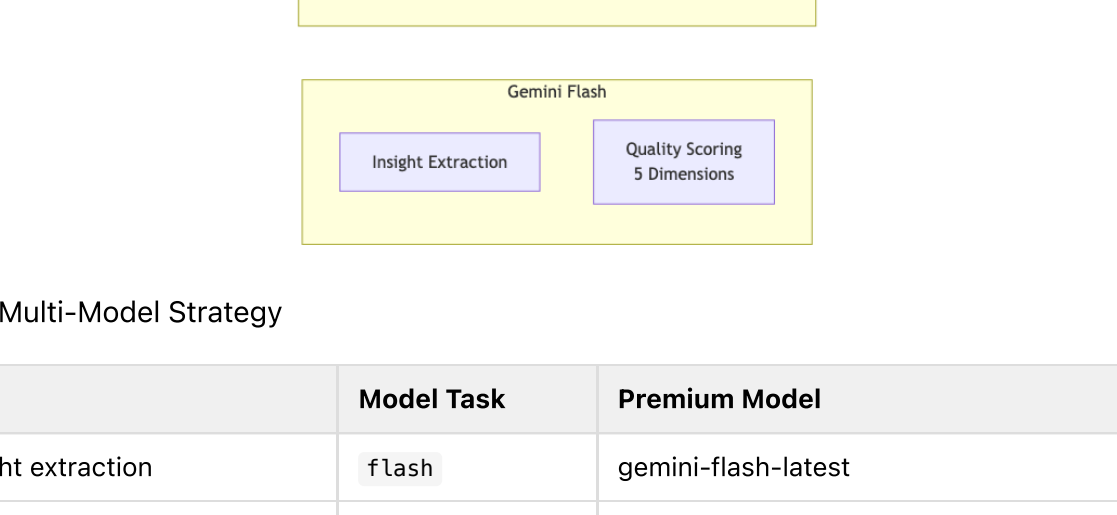
1. **Slop** — Pattern-based + AI analysis (lower is better)
2. **Vendor-Speak** — Vendor language detection (lower is better)
3. **Authenticity** — Human-likeness scoring (higher is better)
4. **Specificity** — Concrete detail scoring (higher is better)
5. **Persona-Fit** — Target audience resonance (higher is better)

Quality gates: Per-voice-profile thresholds. Defaults: `slopMax: 5`, `vendorSpeakMax: 5`, `authenticityMin: 6`, `specificityMin: 6`, `personaMin: 6`

Refinement Loop:

1. Score content
2. If ≥ 2 scorers failed $\rightarrow$ skip refinement, mark for manual review
3. For up to N iterations (default 3): deslop if needed $\rightarrow$ refine from failing scores $\rightarrow$ re-score $\rightarrow$ stop if plateau
4. Return best content + scores

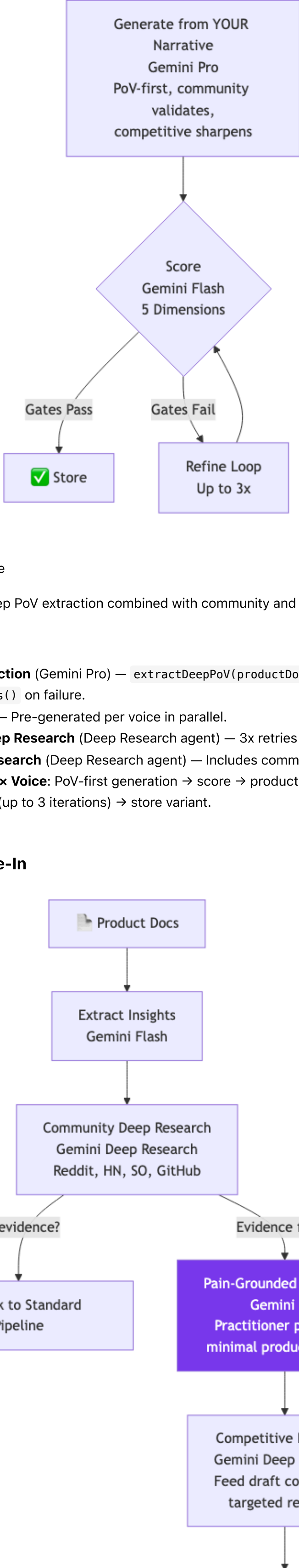
Multi-Model Strategy



Multi-Model Strategy

Step	Model Task	Premium Model
Insight extraction	flash	gemini-flash-latest
Deep PoV extraction	pro	gemini-pro-latest
Session naming	flash	gemini-flash-latest
Banned words generation	flash	gemini-flash-latest
Community research	deepResearch	deep-research-max-preview-04-2026
Competitive research	deepResearch	deep-research-max-preview-04-2026
Content generation	pro	gemini-pro-latest
Attack prompts	pro	gemini-pro-latest
Scoring	all 5 scorers	gemini-flash-latest
Deslop	deslop	gemini-pro-latest

Pipeline 1: Standard



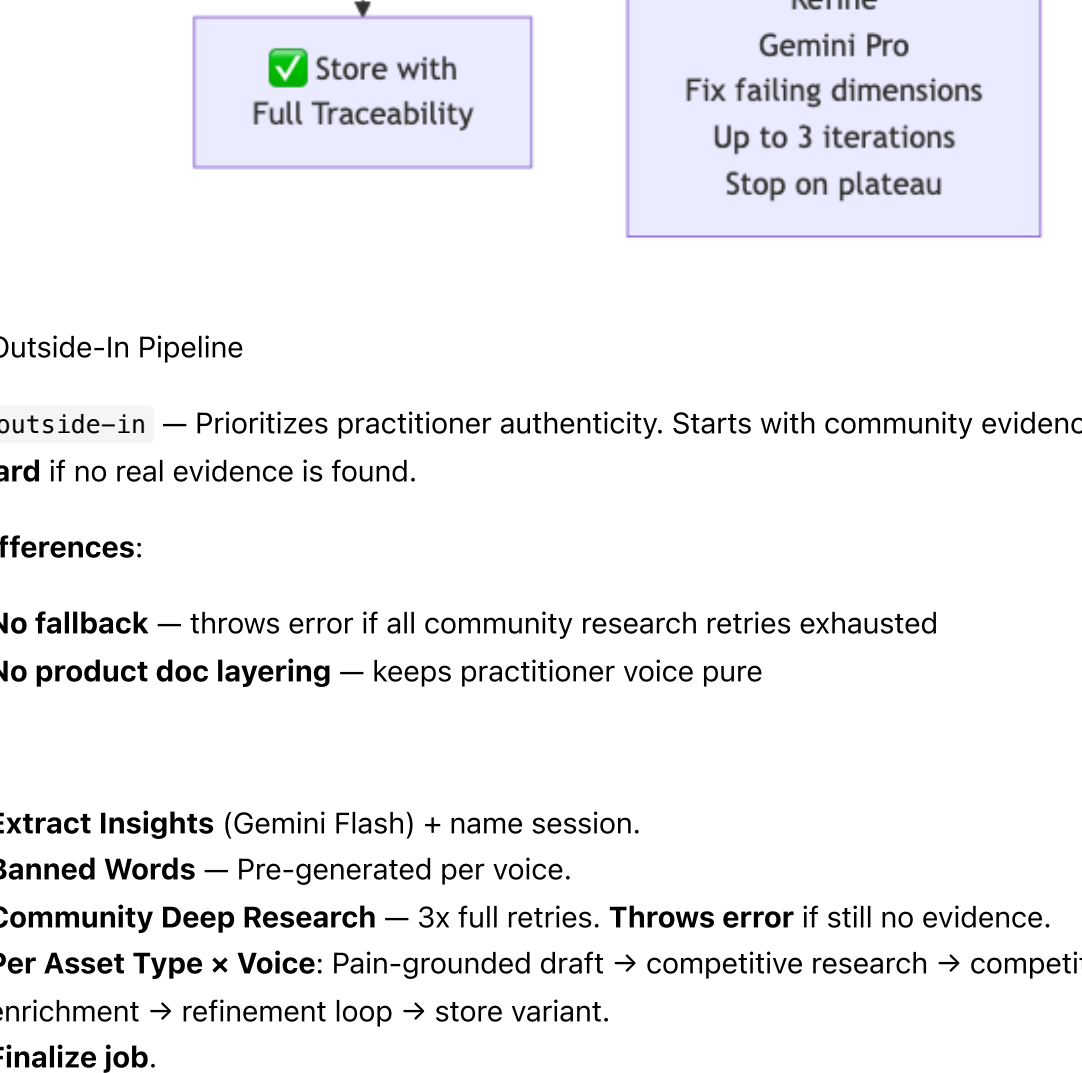
Standard Pipeline

Slug: `standard` — Deep PoV extraction combined with community and competitive research.

Steps:

1. **Deep PoV Extraction** (Gemini Pro) — `extractDeepPoV(productDocs)` . Falls back to `extractInsights()` on failure.
2. **Banned Words** — Pre-generated per voice in parallel.
3. **Community Deep Research** (Deep Research agent) — 3x retries if no evidence.
4. **Competitive Research** (Deep Research agent) — Includes community findings.
5. **Per Asset Type x Voice:** PoV-first generation $\rightarrow$ score $\rightarrow$ product doc layering $\rightarrow$ refinement loop (up to 3 iterations) $\rightarrow$ store variant.
6. **Finalize job.**

Pipeline 2: Outside-In



Outside-In Pipeline

Slug: `outside-in` — Prioritizes practitioner authenticity. Starts with community evidence and **fails hard** if no real evidence is found.

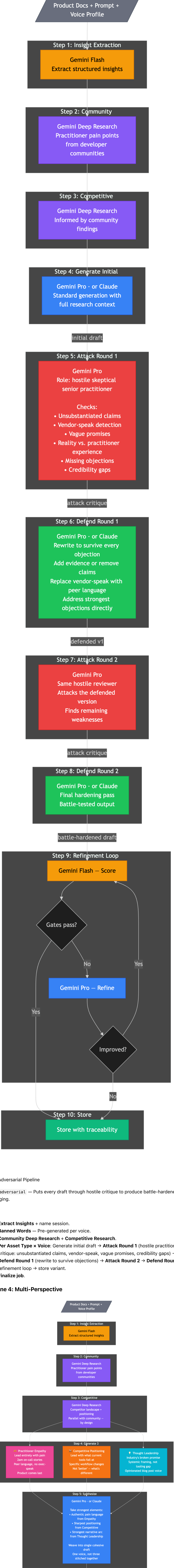
Key Differences:

- **No fallback** — throws error if all community research retries exhausted
- **No product doc layering** — keeps practitioner voice pure

Steps:

1. **Extract Insights** (Gemini Flash) + name session.
2. **Banned Words** — Pre-generated per voice.
3. **Community Deep Research** — 3x full retries. **Throws error** if still no evidence.
4. **Per Asset Type x Voice:** Pain-grounded draft $\rightarrow$ competitive research $\rightarrow$ competitive enrichment $\rightarrow$ refinement loop $\rightarrow$ store variant.
5. **Finalize job.**

Pipeline 3: Adversarial



Adversarial Pipeline

Slug: adversarial — Puts every draft through hostile critique to produce battle-hardened messaging.

Steps:

1. **Extract Insights** + name session.
2. **Banned Words** — Pre-generated per voice.
3. **Community Deep Research** + **Competitive Research**.
4. **Per Asset Type x Voice:** Generate initial draft → **Attack Round 1** (hostile practitioner critique: Unsubstantiated claims, vendor-speak, vague promises, credibility gaps) → **Defend Round 1** (rewrite to survive objections) → **Attack Round 2** → **Defend Round 2** → refinement loop → store variant.
5. **Finalize job**.

Pipeline 4: Multi-Perspective

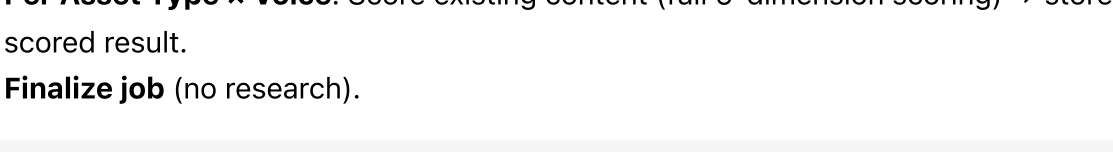
Multi-Perspective Pipeline

Slug: multi-perspective — Generates from 3 angles and synthesizes the best elements.

Steps:

1. **Extract Insights** + name session.
2. **Banned Words** — Pre-generated per voice.
3. **Community Deep Research** + **Competitive Research**.
4. **Per Asset Type x Voice:** Generate initial draft → **3 parallel perspective rewrites** (Empathy, Competitive, Thought Leadership) → **Synthesize** best elements → **Score all 4** (3 perspectives + synthesis), keep highest → refinement loop → store variant.
5. **Finalize job**.

Pipeline 5: Straight-Through



Straight-Through Pipeline

Slug: straight-through — Scores existing content without generating new content.

Prerequisites: Requires `existingMessaging` content in session metadata.

Steps:

1. **Extract Insights** + name session.
2. **Per Asset Type x Voice:** Score existing content (full 5-dimension scoring) → store scored result.
3. **Finalize job** (no research).

```
graph TD
    A[Import Existing Content] --> B[Extract Product Insights]
    B --> C[Score Content - 5 Dimensions]
    C --> D{Passes Quality Gates?}
    D -- Yes --> E[Store with Scores]
    D -- No --> A
    E --> F[Results in Workspace]
```

style A fill:#e1f5fe,stroke:#0288d1
style B fill:#f3e5f5,stroke:#7b1fa2
style C fill:#fff3e0,stroke:#ef6c00
style D fill:#fce4ec,stroke:#c62828
style E fill:#e8f5e9,stroke:#2e7d32
style F fill:#f5f5f5,stroke:#616161

```
subgraph "No Generation / No Transformation"
    A
    B
    C
    D
    E
    F
end
```

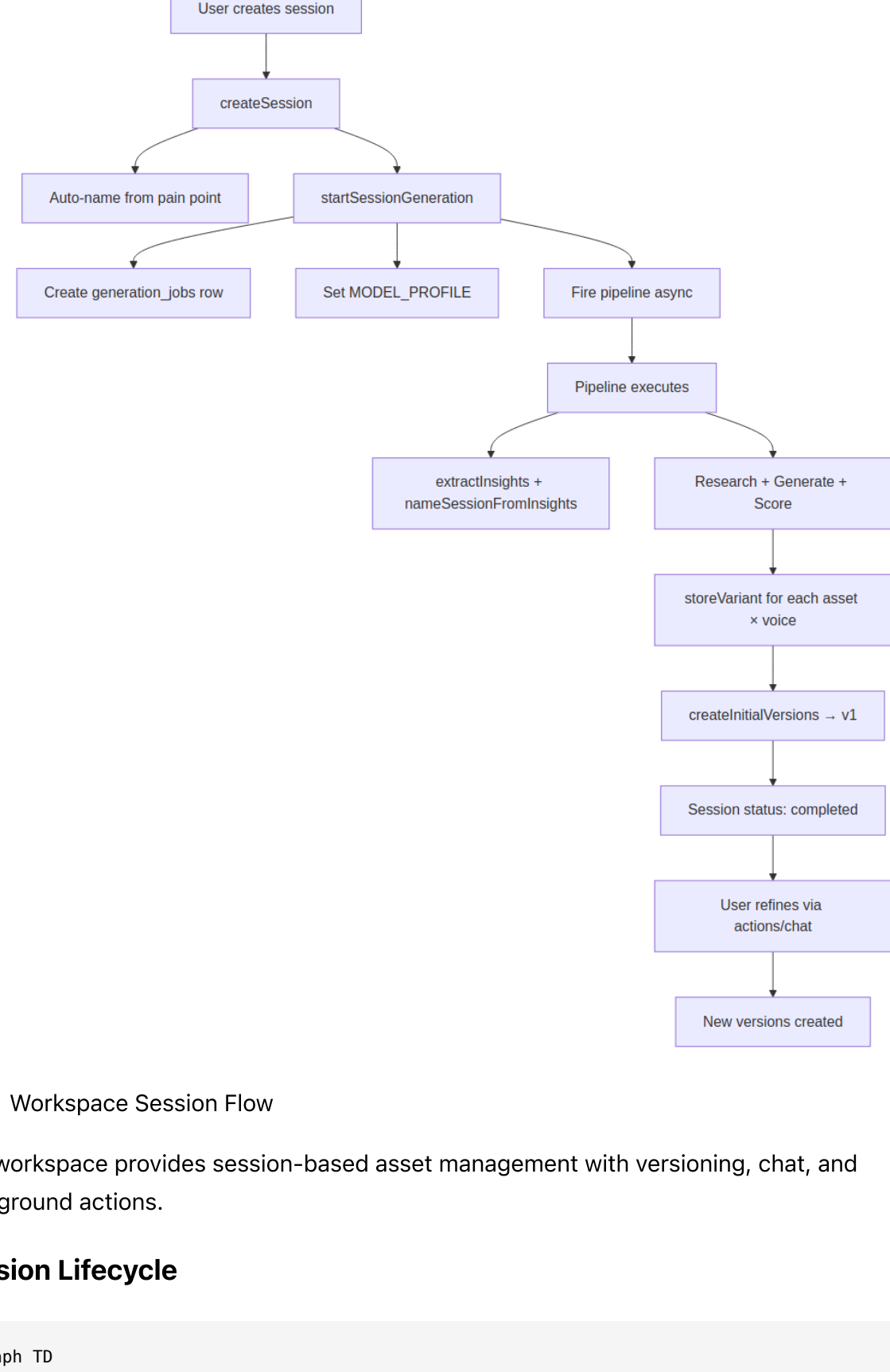
Workspace Actions

Post-generation operations that create new versions within a session, using the same underlying primitives.

Action	Function	Description
Deslop	<code>runDeslopAction</code>	Analyze slop → deslop → score → new version
Regenerate	<code>runRegenerateAction</code>	Full regeneration with voice + template + refinement
Voice Change	<code>runVoiceChangeAction</code>	Rewrite in different voice profile
Adversarial Loop	<code>runAdversarialLoopAction</code>	1–3 iterations; fix mode or elevation mode
Competitive Deep Dive	<code>runCompetitiveDeepDiveAction</code>	Deep Research → competitive enrichment
Community Check	<code>runCommunityCheckAction</code>	Deep Research → practitioner language rewrite
Multi-Perspective	<code>runMultiPerspectiveAction</code>	3 angles → synthesize → score all 4, keep best

All actions execute via `runActionInBackground()` with `action_jobs` tracking. Frontend polls status every 3s.

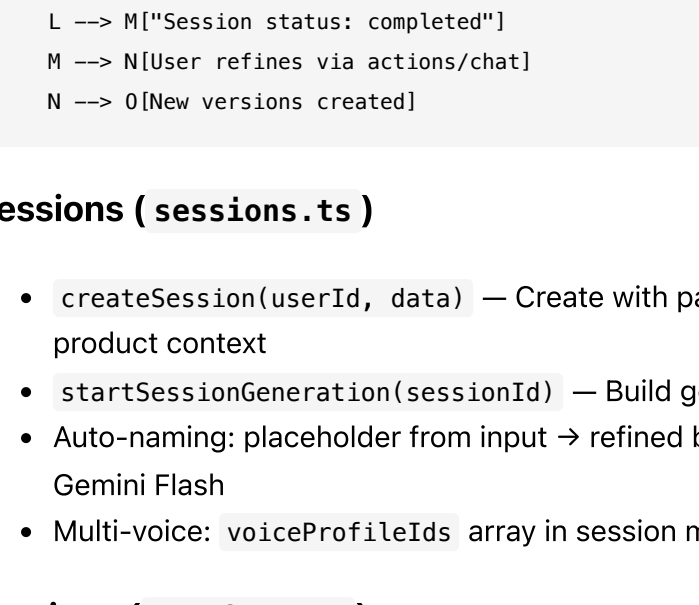
5. Workspace & Sessions



Workspace Session Flow

The workspace provides session-based asset management with versioning, chat, and background actions.

Session Lifecycle



Sessions (sessions.ts)

- `createSession(userId, data)` — Create with pain point, voice(s), asset types, pipeline, product context
- `startSessionGeneration(sessionId)` — Build generation job, fire pipeline in background
- Auto-naming: placeholder from input → refined by `nameSessionFromInsights()` via Gemini Flash
- Multi-voice: `voiceProfileIds` array in session metadata

Versions (versions.ts)

- `createInitialVersions(sessionId, jobId)` — Copy generation results as v1
- `createEditVersion(sessionId, assetType, content)` — User inline edit → score → new version
- `activateVersion(sessionId, versionId)` — Switch active version (deactivates others)
- Every version is scored and stores `isActive` flag

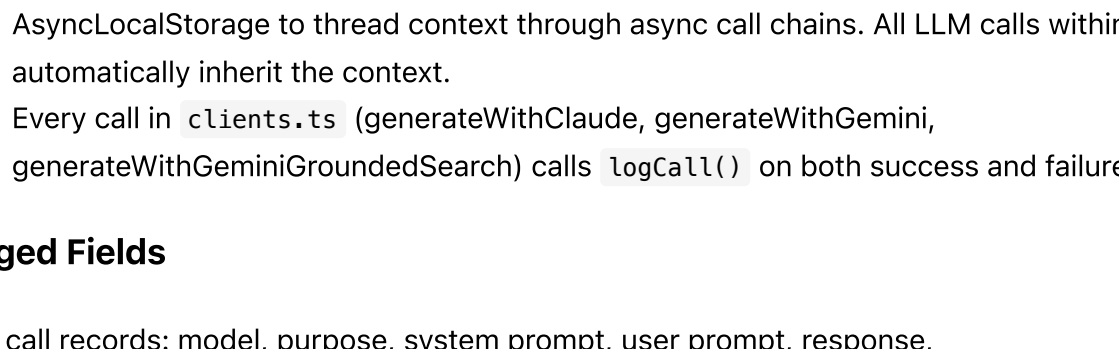
Chat (chat-context.ts)

- `assembleChatContext(sessionId, assetType?)` — Builds system prompt + message history
- System prompt includes: voice guide, anti-slop rules, product context, active version content, other asset summaries
- 150K token budget, oldest-first trimming
- Proposed content wrapped in `-----PROPOSED-----` delimiters for accept/reject

Action Runner (action-runner.ts)

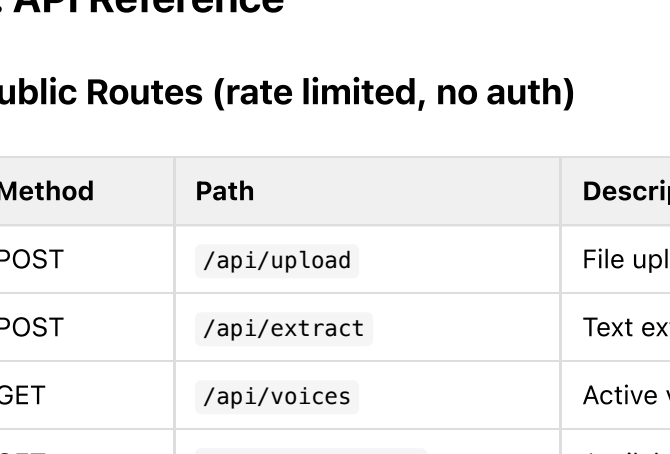
- `runActionInBackground(sessionId, assetType, actionName, actionFn)` — Fire-and-forget with `action_jobs` tracking
- `getActionJobStatus(jobId)` — Poll progress

6. LLM Call Logging



LLM Call Logging

Every AI call is logged to the `llm_calls` table for full auditability.



Components

- `call-logger.ts` — `logCall()` persists every LLM call. Fire-and-forget, never throws.
- `call-context.ts` — `withLLMContext({purpose, jobId, sessionId}, fn)` uses `AsyncLocalStorage` to thread context through async call chains. All LLM calls within `fn` automatically inherit the context.
- Every call in `clients.ts` (`generateWithClaude`, `generateWithGemini`, `generateWithGeminiGroundedSearch`) calls `logCall()` on both success and failure.

Logged Fields

Each call records: model, purpose, system prompt, user prompt, response, input/output/total/cached tokens, latency in ms, success flag, error message, finish reason, linked session ID, linked job ID, and timestamp.

7. API Reference

Public Routes (rate limited, no auth)

Method	Path	Description
POST	<code>/api/upload</code>	File upload
POST	<code>/api/extract</code>	Text extraction from uploaded files
GET	<code>/api/voices</code>	Active voice profiles
GET	<code>/api/asset-types</code>	Available asset types
GET	<code>/api/history</code>	Past generations
POST	<code>/api/auth/login</code>	Login (users table + env var fallback)
POST	<code>/api/auth/signup</code>	User registration

Admin Routes (JWT auth required)

Method	Path	Description
*	<code>/api/admin/documents</code>	Product document CRUD
*	<code>/api/admin/voices</code>	Voice profile CRUD
*	<code>/api/admin/settings</code>	Settings management
GET	<code>/api/admin/stats</code>	Dashboard statistics

Workspace Routes (JWT auth required)

Method	Path	Description
GET	<code>/api/workspace/sessions</code>	List user sessions
POST	<code>/api/workspace/sessions</code>	Create + start session
GET	<code>/api/workspace/sessions/:id</code>	Get session with results/versions
GET	<code>/api/workspace/sessions/:id/status</code>	Poll generation progress
PATCH	<code>/api/workspace/sessions/:id</code>	Update session (name, archive)
DELETE	<code>/api/workspace/sessions/:id</code>	Delete session
GET	<code>/api/workspace/sessions/:id/versions/:assetType</code>	List versions
POST	<code>/api/workspace/sessions/:id/versions/:assetType/edit</code>	Create edit version
POST	<code>/api/workspace/sessions/:id/versions/:versionId/activate</code>	Activate version
POST	<code>/api/workspace/sessions/:id/actions/:assetType/:action</code>	Run workspace action
GET	<code>/api/workspace/sessions/:id/actions/:jobId</code>	Poll action progress
POST	<code>/api/workspace/sessions/:id/chat</code>	SSE streaming chat refinement
GET	<code>/api/workspace/sessions/:id/messages</code>	Chat message history
GET	<code>/api/workspace/sessions/:id/llm-calls</code>	LLM call log for session

8. Design Decisions & Evolution

This section captures key design decisions and the reasoning behind them, drawn from the project's commit history. Understanding *why* things are the way they are prevents repeating past mistakes.

The God File Extraction

`src/api/generate.ts` started as a monolithic “god file” containing all pipeline logic, prompt builders, evidence gathering, and variant storage. It was extracted into `src/services/pipeline/` modules. The key insight: **single source of truth for generate+score+refine**, reused by both pipelines and workspace actions.

Authenticity Scoring Was Faked

A critical bug: `scoreContent()` was copy-pasted into 3 locations, and 2 of them faked the authenticity score as `Math.max(0, 10 - vendorSpeakScore)` instead of calling the real scorer. Fixed by creating a single shared `score-content.ts` module. **Lesson:** never duplicate scoring logic.

Outside-In: Fail Hard, No Fallback

The outside-in pipeline originally fell back to the standard pipeline when no community evidence was found, silently producing mislabeled output. Fix: **fail hard with an explicit error**. If the user wants standard pipeline behavior, they should select the standard pipeline.

Product Doc Layering Removed from Outside-In

Step 6 of the outside-in pipeline (“layer product specifics”) caused product docs to override the practitioner voice. It was removed entirely, keeping practitioner pain front and center.

Evidence Grounding: 3 Generations

- Direct API scrapers** (Reddit, HN, SO, GitHub, Discourse) — fragile, rate-limited
- Gemini grounded search** — AI-powered but keyword-based, flaky
- Gemini Deep Research** — single call searches all sources natively

The scrapers were deleted (~1,109 lines). Deep Research is now the sole evidence source.

Domain-Agnostic Prompts

All prompts originally had hardcoded vertical and practitioner language. Replaced with domain-agnostic language inferring the product domain from extracted insights. The engine now works for any product domain.

Dynamic Banned Words

Evolved from static hardcoded list → dynamic per-voice generation via Gemini Flash. LLM generates 15–20 voice/domain-specific banned words, cached per `voiceId:domain`. Static list kept as fallback (3x retry with backoff).

maxTokens Truncation Bugs

Hardcoded `maxTokens: 50` on session naming caused 40% empty responses. Fix: **never hardcode maxTokens unless you need more than the default**.

Async Background Jobs

All 7 workspace actions originally ran synchronously, causing Cloudflare 100s timeout. Solution: `action_jobs` table with fire-and-forget execution. Frontend polls every 3s.

PM2: tsx watch → node dist

PM2 with `tsx watch` caused EADDRINUSE crashes during in-flight Deep Research. Switched to `node dist/index.js` for stability.

Chat Switched from Claude to Gemini

Workspace chat switched from Claude to Gemini Pro to keep the system on a single AI provider by default. Claude remains available as opt-in override.

Model Profile System

Prevents accidental premium-model spend during testing. `MODEL_PROFILE=economy` uses cheaper latest Gemini aliases. A guard test fails if tests run against premium models.

Sequential DAG Pipelines

Pipelines refactored from parallel to sequential DAGs — each step feeds the next. Only multi-perspective retains parallel generation (3 perspectives) by design.

Tiered Insights Replace Raw Truncation

Every pipeline was independently truncating raw product docs. Replaced with `extractInsights()` running once per job and 4 tiered formatters providing the right level of context to each stage.

LLM-Based Spirit Validation

E2E tests use Gemini to score whether the output matches the pipeline's *intent* (e.g., outside-in must be practitioner-driven, adversarial must feel battle-tested).

No Cron

There is no scheduler/cron. All work is triggered by API requests. A unit test enforces this by scanning source files.

Dead Code Sweep

A major cleanup deleted 25 dead files: 7 frontend pages, 6 admin routes, 11 services, and 18 dead API client methods. The discovery/scheduling/cron pipeline was replaced by the on-demand workspace model.

9. Development Guide

Environment Setup

Key Environment Variables

Variable	Required	Default	Description
PORT	No	3007	API server port
DATABASE_URL	No	./data/messaging-engine.db	SQLite database path
GOOGLE_AI_API_KEY	Yes	—	Google AI API key
ANTHROPIC_API_KEY	Yes	—	Anthropic API key
JWT_SECRET	Yes	—	JWT signing secret
ADMIN_USERNAME	Yes	—	Default admin username
ADMIN_PASSWORD	Yes	—	Default admin password
MODEL_PROFILE	No	production	'economy' or 'premium'
NODE_ENV	No	development	Environment mode

Testing

- Framework:** Vitest with 5-minute timeout per test
- Pool:** forks (single fork — SQLite isn't thread-safe)

- **Model profile:** Tests run with `MODEL_PROFILE=economy` (all Gemini 2.5 Flash)

Test Categories:

- **E2E** (`tests/e2e/`) — Pipeline end-to-end, all 5 pipelines, community evidence, spirit scoring
- **Unit** (`tests/unit/`) — Architecture guards, auth, product insights, quality scoring, workspace operations
- **Integration** (`tests/integration/`) — Session naming with real Gemini API

Common Tasks

Adding a New Asset Type:

1. Create template in `templates/`
2. Add to `ALL_ASSET_TYPES` array in `src/services/pipeline/prompts.ts`
3. Add entry in `ASSET_TYPE_LABELS` and `ASSET_TYPE_TEMPERATURE`
4. Add type to the `AssetType` union in `src/services/generation/types.ts`

Adding a New Pipeline:

1. Create pipeline file in `src/services/pipeline/pipelines/`
2. Export `runMyPipeline(jobId, inputs)` function
3. Register in `PIPELINE_RUNNERS` map in `orchestrator.ts`
4. Compose from shared primitives

Adding a New Workspace Action:

1. Add action function in `src/services/workspace/actions.ts`
2. Use `withLLMContext()` for automatic call logging
3. Use `createVersionAndActivate()` to store results
4. Register in action dispatch in `src/api/workspace/sessions.ts`

Hard-Won Rules

These patterns are learned from the project's evolution. Violating them has caused real bugs.

1. **Never duplicate `scoreContent()`** — Authenticity scoring was faked in 2 of 3 copy-pasted locations. Always import from `score-content.ts`.
2. **Never hardcode `maxTokens` low** — `maxTokens: 50` caused 40% empty responses. Only set when you need more than the 8192 default.
3. **Outside-In must fail hard** — No silent fallback to standard pipeline. Throw error if no evidence after retries.
4. **Product docs override practitioner voice** — Be cautious about mixing product positioning into practitioner-first content.
5. **Grounded search is flaky** — Returns 200 OK with 0 results non-deterministically. Always retry (5x).
6. **Domain-agnostic prompts only** — No hardcoded domain language. All context from extracted insights.
7. **No file watchers in production** — PM2 with `tsx watch` causes EADDRINUSE crashes during Deep Research.
8. **Compose, don't duplicate** — Workspace actions must compose from pipeline primitives. Exception: `runAdversarialLoopAction` (unique elevation mode).
9. **Evidence retries are layered** — Grounded search: 5x. Community deep research: 3x. Separate retry loops at different levels.
10. **Test profile guard** — `model-profile-guard.test.ts` fails if tests hit production models.

Key File Locations

<code>src/</code>	
<code> index.ts</code>	<code># Server entry</code>
<code> config.ts</code>	<code># Config + Model Profile System</code>
<code> db/</code>	
<code>schema.ts</code>	<code># All 20 table definitions</code>
<code>index.ts</code>	<code># Database connection</code>
<code> services/</code>	
<code>ai/</code>	
<code>clients.ts</code>	<code># Gemini + Claude clients</code>
<code>call-logger.ts</code>	<code># Fire-and-forget LLM call logging</code>
<code>call-context.ts</code>	<code># AsyncLocalStorage context threading</code>
<code>auth/</code>	
<code>users.ts</code>	<code># User registration/authentication</code>
<code>pipeline/</code>	
<code>orchestrator.ts</code>	<code># Shared primitives + dispatch</code>
<code>evidence.ts</code>	<code># Community/competitive research</code>
<code>prompts.ts</code>	<code># All prompt builders + templates</code>
<code>pipelines/</code>	
<code>standard.ts</code>	
<code>outside-in.ts</code>	
<code>adversarial.ts</code>	
<code>multi-perspective.ts</code>	
<code>straight-through.ts</code>	
<code>workspace/</code>	
<code>sessions.ts</code>	<code># Session CRUD + generation kickoff</code>
<code>versions.ts</code>	<code># Version management</code>
<code>actions.ts</code>	<code># 7 workspace actions</code>
<code>action-runner.ts</code>	<code># Background action execution</code>
<code>chat-context.ts</code>	<code># Chat context assembly</code>
<code>quality/</code>	
<code>score-content.ts</code>	<code># Centralized 5-scorer system</code>
<code>slop-detector.ts</code>	<code># Pattern detection + AI analysis</code>
<code>grounding-validator.ts</code>	<code># Fabrication detection</code>
<code>product/</code>	
<code>insights.ts</code>	<code># Product insight extraction</code>
<code>admin/</code>	<code># Vite + React workspace UI</code>
<code>templates/</code>	<code># 8 asset type prompt templates</code>
<code>tests/</code>	
<code>e2e/</code>	<code># End-to-end pipeline tests</code>
<code>unit/</code>	<code># Unit tests</code>
<code>integration/</code>	<code># Integration tests</code>